



Rejoy Chakraborty

Ph.D. Scholar

Computer Vision and Pattern Recognition Unit

Indian Statistical Institute, Kolkata

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EDUCATION

Degree	Institute	Subject	CGPA/Percentage	Year
M.Tech.	Indian Institute of Technology Ropar (IIT Ropar)	CSE	9.6	2023-2025
M.Sc.	West Bengal State University	CS	9.7	2021-2023
B.Sc. (Hons.)	Ramakrishna Mission Vivekananda Centenary College	CS	9.33	2018-2021
Higher Secondary	Bidhannagar Municipal School	Science	73.8%	2018

AREA OF INTEREST

- **Primary Interest:** Generative AI, Computer Vision
- **Secondary Interest:** Few Shot Learning, Deep Learning

RESEARCH EXPERIENCE

- **Indian Statistical Institute, Kolkata** *July 2025 - Present*
Junior Research Fellow (JRF)
Kolkata, West Bengal
 - **Supervisor:** Prof. Umapada Pal
 - Joined as ISI Research Scholar at CVPR Unit
- **University of Calcutta** *May 2019 - June 2019*
Summer Intern
Kolkata, West Bengal
 - **Supervisor:** Prof. Samir Kumar Bandyopadhyay
 - Worked on Audio Steganography.

TEACHING EXPERIENCE

- **Indian Statistical Institute, Kolkata** *August 2025 - December 2025*
Teaching Assistant (TA)
Kolkata, West Bengal
 - Introduction to Programming
- **Indian Institute of Technology Ropar** *July 2024 - May 2025*
Teaching Assistant (Head TA)
Rupnagar, Punjab
 - Data Structures and Algorithms (CS506 & CS205)
 - Digital Image Processing (CS517)

PUBLICATIONS

- **DRG-Font: Dynamic Reference-Guided Few-shot Font Generation via Contrastive Style-Content Disentanglement** *2026*
Under Review
 - Authors: Rejoy Chakraborty, Prasun Roy, Saumik Bhattacharya, Umapada Pal
 - DOI: <https://doi.org/10.48550/arXiv.2604.13797>
- **Source Camera Model Identification via Federated Learning using Laplacian-based Patches** *2025*
IEEE Transactions on Artificial Intelligence (IEEE TAI)
 - Authors: Rejoy Chakraborty, Puneet Goyal
 - DOI: <https://doi.org/10.1109/TAI.2025.3629009>
- **Camera Model Identification with SPAIR-Swin and Entropy based Non-Homogeneous Patches** *2025*
Preprint
 - Authors: Protayay Dey, Rejoy Chakraborty, Abhilasha S. Jadhav, Kapil Rana, Gaurav Sharma, Puneet Goyal
 - DOI: <https://doi.org/10.48550/arXiv.2503.22120>
- **MSPlantNet: Plant Disease Classification using Few Shot Learning** *2024*
9th International Conference on Computer Vision and Image Processing (CVIP 2024)
 - Authors: Rejoy Chakraborty, Md. Sajid Anis, Pallav Kumar, Manish Jha, Neeraj Goel, Mukesh Saini
 - URL: https://link.springer.com/chapter/10.1007/978-3-031-93697-5_24
- **MSBNet: Handwritten Bangla Character Recognition Using Lightweight Multi-scale CNN Architecture** *2024*
8th ACM India Joint International Conference on Data Science and Management of Data (CODS-COMAD Dec'2024)
 - Authors: Rejoy Chakraborty, Chayan Halder, Kaushik Roy, Shivam Gupta, Shashi Shekhar Jha
 - DOI: <https://doi.org/10.1145/3703323.3703340>

PROJECTS

- **Source Camera Model Identification using Federated Learning** *June 2024 - May 2025*
Supervisor: Dr. Puneet Goyal
 - M.Tech. Thesis Project
 - Source camera model identification from novel curated document images and incorporating Federated Learning for addressing privacy concern.
- **Hand Gesture Recognition using Few Shot Learning** *April 2024 - May 2024*
Instructor : Dr. Santosh K. Vipparthi
 - Hand gesture recognition on HaGRID dataset using Few-shot learning with multi-scale architecture
- **Semi-automated Ground Truth Generation of Bangla Handwritten Offline Text Documents** *Aug 2022 - July 2023*
Supervisor: Prof. Kaushik Roy
 - M.Sc. Thesis Project
 - Designed a semi automated system for line, word and character segmentation and ground truth generation from Bangla handwritten offline text documents
- **Generating Convex Hull From 3D Point Cloud Data Denoised Using Bilateral Filtering** *Jan 2018 - July 2018*
Supervisor: Dr. Biswajit Biswas
 - B.Sc. Thesis Project
 - Applied Bilateral Filtering scheme for outlier removal on Point Cloud data and generated Convex Hull on it.

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, LaTeX, Java
- **Frameworks/Packages:** PyTorch, Tensorflow, OpenCV
- **Operating Systems:** Windows, Linux

SUBJECT SKILLS

- **Core:** Data Structure and Algorithms, Linear Algebra, Machine Learning
- **Others:** Artificial Intelligence, Game Theory

REVIEW EXPERIENCE

- **Reviewer:** IEEE TETCI, ACM TALLIP, ICDAR, CVIP, SNCS *2023-Present*
- **External Reviewer:** ICPR *2026*

ACHIEVEMENTS

- **Institute Silver Medalist**, Topper in M.Tech. in CSE, Indian Institiute of Technology Ropar *2025*
- **Recieved Reliance Foundation Postgraduate Scholarship**, during the tenure of M.Tech. course *2023-2025*
- **Qualified GATE 2023 in CSIT**, with All India Rank 983 and Score 626 *2023*
- **Received Swami Vivekananda Merit-cum-Means Scholarship**, Govt. of West Bengal funded *2021-2023*
- **Gold Medalist**, Topper in B.Sc. CS (Hons), Ramakrishna Mission Vivekananda Centenary College *2021*

VOLUNTEER EXPERIENCE

- **27th International Conference on Pattern Recognition (ICPR 2024)** *December 2024*
Student Volunteer
 - Organized at Biswa Bangla Convention Centre, Kolkata, India
- **2nd International Conference on Agriculture-Centric Computation (ICA 2024)** *May 2024*
Student Volunteer
 - Organized at Indraprastha Institute of Information Technology Delhi, Delhi, India

OTHER INFORMATIONS

- **Date of Birth:** 29/12/1999
- **Marital Status:** Unmarried
- **Soft Skills:** Hard worker, Management, Leadership
- **Hobbies:** Teaching, Singing